FFFA Donor Management MVP — System Design

Version: v1.1 (19 Oct 2026, includes Chris's review) · **Status:** Reviewed by Systems lead · **Source of truth:** PRD v2 (8 Oct 2026) · **Owner:** Systems workstream

This document describes how we build Phase 1 (DM1–DM10) by 28 Oct 2026. Anything not in PRD v2 is out of scope unless the team agrees to add it.

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1. Goals and non-goals

Goals (Phase 1)

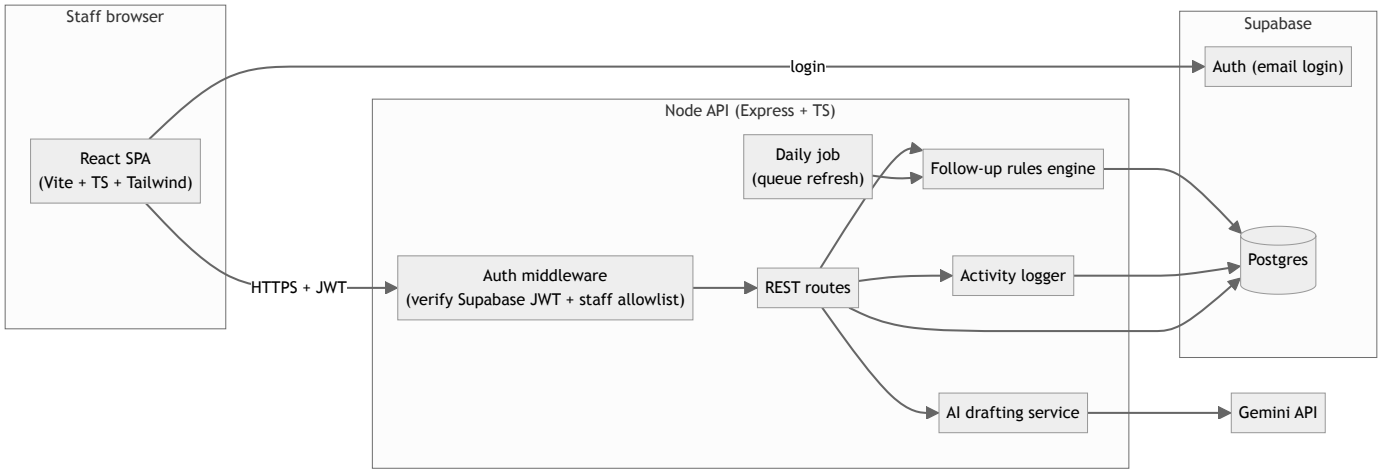
- One place where FFFA staff keep every donor, donation and relationship note (DM1, DM3, DM8).
- Follow-ups surface automatically from agreed rules instead of staff memory (DM9, DM4).
- Staff can generate an AI-drafted, personal follow-up, edit it and approve it; nothing sends automatically (DM10).
- Search, filters, dashboard, CSV import, staff-only access and an activity log (DM2, DM5, DM6, DM7).

Non-goals (Phase 1) — from PRD v2 "Out of scope"

- Sending email/WhatsApp from the system, newsletters, campaign ads, donor portal, payments.
- Volunteer or beneficiary management (only an "also volunteers" tag on donors).
- IRAS submission (but donation records stay structured so a later export is easy).
- NRIC/FIN/UEN or any sensitive attributes (age, income, ethnicity).

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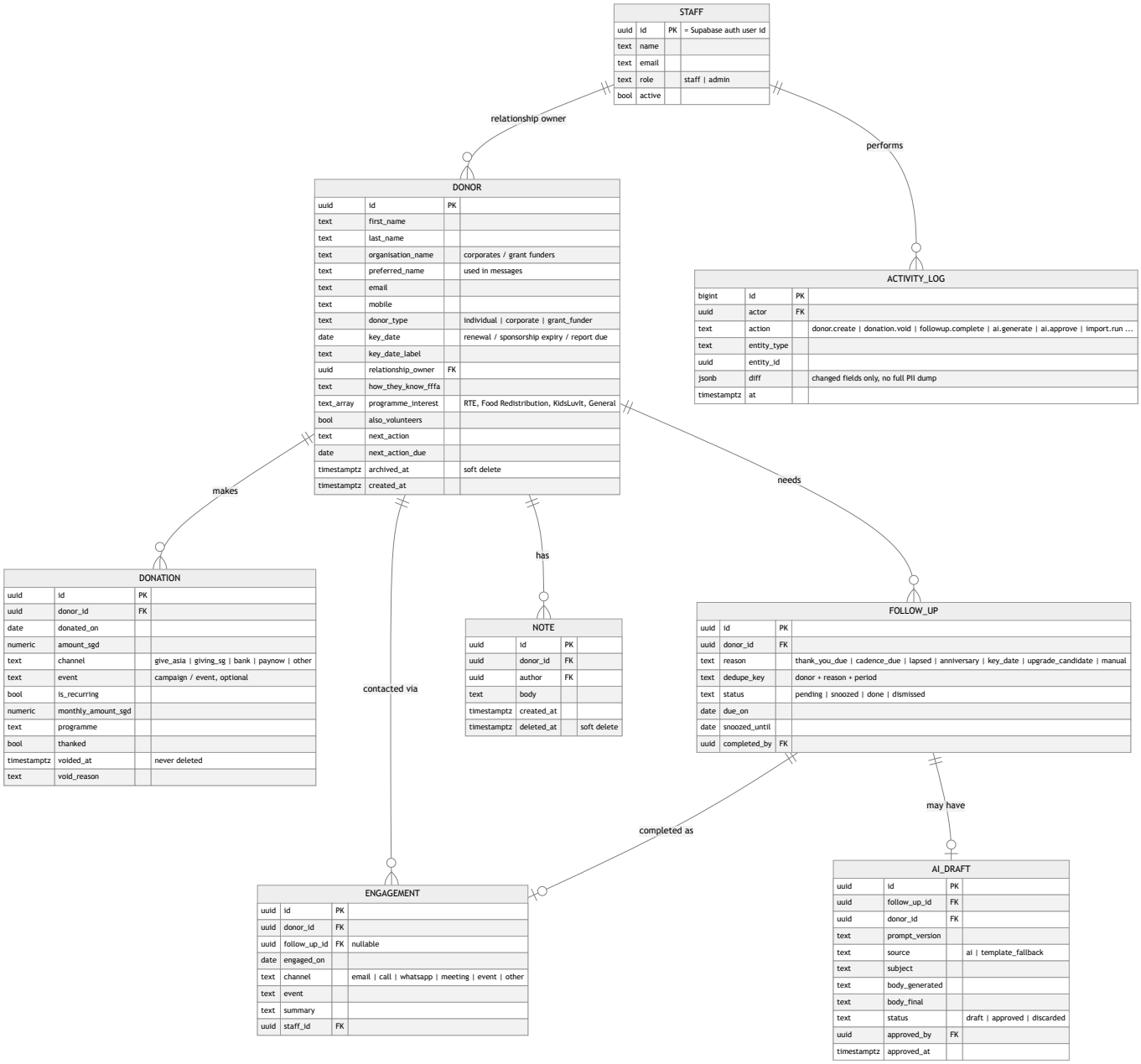
2. Architecture



Key decisions

Decision	Choice	Why	Alternatives considered
Frontend	Keep React 19 + Vite + TS + Tailwind from the current repo	Team already has it; fast to build	Next.js (more setup, no SSR need)
Backend	Keep a thin Express API in front of the DB	Gemini key must stay server-side; rules, logging and AI live in one place; client never touches the DB directly	Client → Supabase directly with Row Level Security (less code, but business rules and audit logging get scattered and the AI key still needs a server)
Database	Supabase Postgres	Relational data (donors ↔ donations ↔ follow-ups), free tier, table editor staff can open in an emergency, backups	Firestore (weaker for joins/aggregates), SQLite (no managed hosting/auth)
Auth	Supabase Auth (email + password or magic link) + server-side staff allowlist	Built-in, no custom password handling	Google Workspace SSO (depends on FFFA's setup — revisit in Phase 2)
AI model	Gemini Flash via `@google/genai` (exact model id verified at build time)	Already wired in repo, cheap, structured JSON output	Claude / GPT (equally viable; switching is a single service file)
Follow-up queue	**Materialised**: rules write `follow_ups` rows; daily job + on-demand refresh	Gives each follow-up a status/history (DM4 "removes from pending"), lets staff snooze/complete, and the dashboard can count them cheaply	Compute on read (simpler, but no status tracking or history)
Hosting	Container on Cloud Run (or Render) — decide with FFFA	FFFA has hosting funding; container keeps us portable	Vercel (fine for SPA, awkward for the daily job)

3. Data model



Computed, never stored — a `donor_summary` view derives these on read so they can't go stale: last donation (date, amount, event), last engagement (date, channel), average donation amount, average days between donations, 12-month giving total, **tier** and **days since last engagement**.

Tiers (internal only). `tier_settings` holds Bronze / Silver / Gold / Diamond with a 12-month giving threshold (e.g. ≥ S$5,000 = Gold) and a **contact cadence in days** per tier. Tier is calculated from 12-month giving, never set by hand. Tiers flagged `personal_outreach` (default: Diamond) are excluded from AI drafting, per PRD "major donors get a human". Tiers are never sent to the AI or shown to donors.

Other config tables: `rule_settings` (thresholds, editable by admin) and `org_settings` (impact unit, e.g. "S$3 = 1 RTE meal", org signature).

Data guardrails (PRD §3): no NRIC/FIN/UEN, no age/income/ethnicity; dummy or de-identified data only until FFFA authorises live data.

4. Follow-up rules engine (DM9)

Runs daily (cron) and on demand ("Refresh queue" button). Each rule is a pure function `(donor, donations, settings, today) → FollowUpCandidate | null`, so rules are easy to unit-test and to add later.

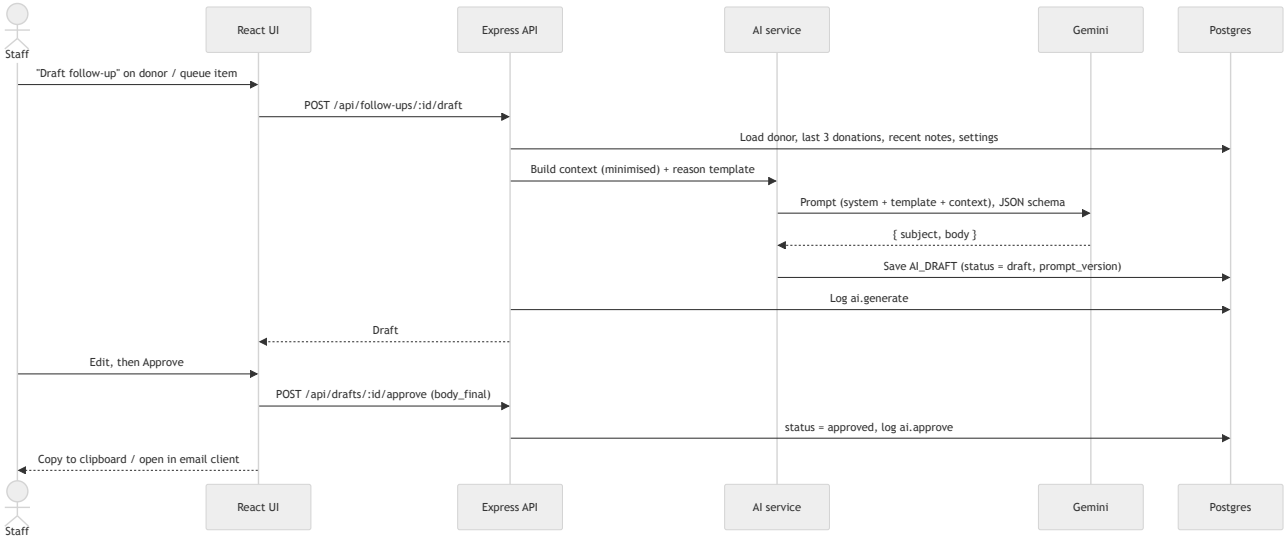
Rule	Fires when	Default threshold (FFFA to confirm)	Priority
`thank_you_due`	A donation has `thanked = false` and is ≥ N days old	N = 3 days	High
`cadence_due`	Days since last engagement ≥ the donor's tier cadence (going cold)	Per tier in `tier_settings`	High
`lapsed`	Recurring donor with no donation for N days, or one-off donor with none for M days	N = 45, M = 365	High
`key_date`	`key_date` (renewal / sponsorship expiry / grant report) within N days	N = 30	High
`anniversary`	First-donation anniversary within the next N days	N = 7	Medium
`upgrade_candidate`	Recurring at the same monthly amount for ≥ 12 months, not asked in the last 12 months	12 months	Low

Idempotency: each candidate has a `dedupe_key` (e.g. `donor:123|lapsed|2026-10`). The engine upserts on that key, so re-running never duplicates and a completed follow-up is not recreated in the same period.

Completing a follow-up records an `ENGAGEMENT` (date, channel, summary), which resets that donor's cadence clock.

Personal-outreach tiers: still appear in the queue, but are tagged "personal outreach" and the AI draft button is disabled for them (PRD: major donors get a human).

5. AI drafting flow (DM10)



- **Prompt templates per reason** (thank-you, lapsed, anniversary, key date, upgrade ask), versioned in code (`prompt_version`) so we can compare outputs.
- **Context minimisation:** the model gets first name, donor type, tenure, donation amounts/dates/programme, relationship notes, the impact unit and the reason. **Never** email, mobile or address.
- **Tone guardrails** in the system prompt: warm, specific, no guilt, upgrade asks modest and concrete ("S$10 → S$15 = X more meals"), Singapore English, FFFA voice.
- **Nothing sends from the system.** Approve = copy or `mailto:`; sending stays with staff (PRD out-of-scope).
- **Failure mode:** if Gemini fails or times out, fall back to the static template for that reason (`source = template_fallback`) so staff are never blocked.

- **Evaluation:** a small fixed set of ~10 dummy donors × 5 reasons; we review outputs for tone, accuracy (no invented facts) and length each time prompts change.

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6. API (v1)

All routes require a valid staff session. Writes go through the activity logger.

Method	Route	PRD
GET	`/api/donors?query=&type=&tier=&lastDonationBefore=&owner=`	DM1, DM2
POST / PATCH	`/api/donors` , `/api/donors/:id`	DM1
GET	`/api/donors/:id` (profile: summary, donations, engagements, notes, follow-ups, drafts)	DM3
POST / DELETE	`/api/donors/:id/notes` , `/api/notes/:id` (soft delete)	DM3
POST	`/api/donors/:id/donations` , `/api/donations/:id/void`	DM8
POST	`/api/donors/:id/engagements`	DM3, DM4
GET	`/api/follow-ups?status=pending` (the "Today" queue)	DM9
POST	`/api/follow-ups/refresh`	DM9
POST	`/api/follow-ups/:id/complete` (creates an engagement) · `/snooze` · `/dismiss`	DM4
POST	`/api/follow-ups/:id/draft` · `/api/drafts/:id/approve` · `/api/drafts/:id/discard`	DM10
GET	`/api/dashboard` (aggregates + monthly series)	DM5
POST	`/api/import/preview` · `/api/import/commit` (CSV)	DM6
GET / PATCH	`/api/settings` (admin: rules, tiers, impact unit)	DM9, DM10
GET	`/api/activity?entity=` (admin only)	DM7
POST / DELETE	`/api/donors/:id/archive` · `DELETE /api/donors/:id` (admin only, PDPA requests)	DM1

Delete semantics. Each resource gets the operation its data needs rather than a blanket DELETE:

Resource	Operation	Why
Donor	Archive (soft); hard delete admin-only	PDPA requests to remove personal data
Donation	Void with reason, never deleted	Financial record; feeds a later IRAS export
Note	Soft delete (author or admin)	Recoverable mistakes
Follow-up	Dismiss	Keeps queue history
AI draft	Discard	Keeps generation history for evaluation
Activity log	Never	Audit trail

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7. Screens

1. **Login**
2. **Today** — pending follow-ups grouped by priority; actions: draft, complete, snooze. *(The demo's home screen.)*
3. **Donors** — table with search and filters.
4. **Donor profile** — details, donations, notes timeline, follow-ups, "Log donation", "Draft follow-up".
5. **Dashboard** — total/active donors, donors by tier, pending follow-ups, upgrade candidates, plus two monthly charts: **donations** and **outbound engagements** (from the engagements table, not the activity log). Aggregates only; built after the core flow.
6. **Import** — upload CSV, preview with validation errors, commit.
7. **Settings** (admin) — rule thresholds, tier thresholds and cadences, impact unit, staff allowlist.

8. Security (DM7, PRD §3)

- Supabase Auth login; the API verifies the JWT on every request **and** checks the user is in the `staff` table and active.
- Roles: `staff` (day-to-day), `admin` (settings, staff list, activity log).
- The browser never gets DB credentials; Postgres Row Level Security set to deny-all for the anon key, server uses the service role.
- Secrets (`GEMINI_API_KEY` , Supabase service key) only in server env.
- Activity log for every write and every AI generate/approve (who, what, when, changed fields).
- Dashboards and exports show aggregates only; no bulk PII export in Phase 1.
- Dummy data in all environments until FFFA signs off on live data.

9. Changes from the current repo

Current (AI Studio demo)	Phase 1 plan
In-memory arrays in `server.ts`	Supabase Postgres
No login	Supabase Auth + staff allowlist
Single 1,200-line `App.tsx`	`src/pages/*` , `src/components/*` , `src/lib/api.ts`
Single `server.ts`	`server/routes/*` , `server/services/{rules,ai,activity}.ts` , `server/db.ts`
Newsletter studio, campaigns, AI insights	Removed from nav (Phase 2 per PRD); code kept on a branch
Generic sample data with real-looking corporate emails	Clearly fictional FFFA-style seed data (RTE, KidsLuvIt, Give Asia, Giving SG)
README describes demo features	Rewrite README to match this design once approved

10. Workstream ownership (Systems)

Estimates are in ideal working days, sized per task so the load is even.

Person	Owns	PRD	Days
Ghanashyam (lead)	Architecture, shared schema/types and repo restructure (1.5) · AI drafting service, prompts and approve flow (2.5) · AI evaluation with Lynn (0.5) · integration, stand-ups, demo prep (1.5)	DM10	~6
Swaraj	Donor, donation and notes API (2) · follow-up rules engine and follow-up API (3) · activity log (1)	DM1–DM4, DM8, DM9, DM7 (log)	~6
Chris	Auth and staff access (1.5) · screens: Login, Today, Donors, Donor profile, Import, Settings (5)	DM7 (auth), DM1–DM4 (UI)	~6.5
Lynn	Seed data (0.5) · CSV import back end (1.5) · dashboard API and screen (1.5) · AI evaluation set (0.5) · test checklist (1)	DM5, DM6, DM10 (eval)	~5

Ways of working

- **Day 1, together:** agree the shared schema, TypeScript types and API conventions before building, so everyone can work in parallel.
- **Peer code review:** every pull request is reviewed by one other teammate; reviewers rotate so everyone understands every part of the system.
- **Short stand-ups** (async in the group is fine): done, next, blocked.
- **Team demos:** the lead opens and closes; each owner presents the part they built.

People & Processes (Fiona's team) provides: FFFA field confirmation, rule thresholds, impact unit, tone guidance and tester names.

11. Milestones

Date	Milestone
19 Oct	Design v1.1; platform foundation (schema, types, DB layer, auth hook, activity log hook, AI service)
20 Oct	Donor API, rules engine core, Today + profile screens on seed data; AI draft → approve working
21 Oct	**Session 2: team demo to mentor of Today → profile → AI draft → approve**
24 Oct	Engagements, tiers, dashboard (with charts), import, settings; FFFA testing
27 Oct	Fixes, staff guide, handover notes
28 Oct	**Final demo and handover**

12. Assumptions and open questions

Assumptions (we build on these; FFFA can correct them anytime)

- Donor fields: as listed in section 3, built and tested on dummy data until FFFA shares a template or de-identified sample.
- Rule thresholds: the defaults in section 4, editable by an admin in Settings.
- Impact unit: a placeholder (e.g. "S$X = 1 RTE meal") until FFFA confirms the figure.
- AI drafts: any staff user can approve; major donors are excluded from AI drafting.
- Email: staff send from their own email client; nothing sends from the system.
- Hosting and post-handover ownership: decided with FFFA before 28 Oct; the build stays container-portable.

Open questions (already sent to FFFA in the question sheet)

1. **Do you hold any other donor data, such as demographics (age, income, ethnicity, dialect) or preferences?** FFFA's stated purpose is clarity and comparison. *Design note:* PRD §3 keeps these fields out of Phase 1; if FFFA has them, we revisit as a Phase 2 decision with privacy controls.
2. **Does your team use a dashboard for key operational figures (active donors, active beneficiaries, inventory availability, etc.)?** *Design note:* the Phase 1 dashboard (DM5) covers donor figures only; beneficiaries and inventory stay out of scope, but the answer tells us what staff are used to seeing.